

Azmat Ullah



[LinkedIn](#)



azmatullah21@ku.edu.tr



[Google Scholar](#)

Education

Koç Üniversitesi, Istanbul, Turkey

September-2021 to July-2026

PhD, Biomedical Engineering

National University of Science and Technology, Islamabad

September-2016 to July-2019

Masters, Biomedical Engineering

University of Engineering and Technology, Peshawar

August-2012 to July-2016

Bachelors, Mechatronics Engineering

Publications

- Ullah, Azmat, Tansu Gölcez, Oğuz Orhun Teber, İsmail Koyuncu, A. Kerem Uğuz, and Kerem Pekkan. "A microfluidic technique for measuring fiber-level mass transport efficiency and hemolysis of blood oxygenators." *Biomicrofluidics* 19, no. 6 (2025).
- Awais, Muhammad, Easa AliAbbasi, Abdulkadir Yasin Atik, Mohammad Javad Bathaei, Mohsin Ali, Ritu Das, Cagdas Dag et al. "Multiplexed Piezoelectric Electronic Skin with Haptic Feedback for Upper Limb Prosthesis." *Advanced Sensor Research* 3, no. 12 (2024): 2400100.
- Khan, Faiz Muhammad, Naila Bibi, Saleem Abdullah, and Azmat Ullah. "Complex fuzzy rough aggregation operators and their applications in for multi-criteria group decision-making." *International Journal of Fuzzy Logic and Intelligent Systems* 23, no. 3 (2023): 270-293.
- AliAbbasi, Easa, Anil Akseki, Azmat Ullah, and Kerem Pekkan. "Tension controlled hollow-fiber winding machine for blood oxygenator prototypes." *HardwareX* 14 (2023): e00434.
- Salman Khan, Muhammad, Azmat Ullah, Kaleem Nawaz Khan, Huma Riaz, Yasar Mehmood Yousafzai, Tawsifur Rahman, Muhammad EH Chowdhury, and Saad Bin Abul Kashem. "Deep learning assisted automated assessment of thalassaemia from haemoglobin electrophoresis images." *Diagnostics* 12, no. 10 (2022): 2405.
- Afaq Khan, Azmat Ullah, Kaleem Nawaz and Muhammad Salman Khan "Automated Assessment of Lymphocytes Using Machine Learning Techniques" *International Conference on Artificial Intelligence*, 2021.
- Muhammad Tahir Khan, Anam Abid, Azmat Ullah, Mansoor alam, Muhammad Sohail, "Real time health monitoring of industrial machine using multiclass support vector machine," 2017 2nd International Conference on Control and Robotics Engineering (ICCRE), Bangkok, 2017, pp. 77-81.
- Azmat Ullah, Ali Madayen, Sercin Ozkok, Kerem Pekkan, "Enhancing Resolution and Cross-Directional Velocity Prediction in Fontan Circulation Phase Contrast MRI Using Physics-Informed Neural Networks" (Under review)
- Azmat Ullah, Kerem Pekkan, "A Low-Cost Multi-Channel Data Acquisition System for Pressure and Flow Measurements" (Under review)

Research Experience

National University of Science and Technology

Islamabad, Pakistan

Research Assistant

March-2017 to February-2018

Projects

- Designed and fabricated data acquisition system for mental stress analysis.

- Participated in the development of experimental setup for Trans-cranial Direct Current Stimulation(tDCs).
- Facilitated preprocessing of dynamic causal modeling (DCM) of Effective Connectivity of Brain Regions.
- Web interface for Wireless Trans-cranial Direct Current Stimulation of mice using ESP8266.
- Distal radius fracture detection in X-ray images using Hugh Transform.
- Design and fabrication of sensory feedback system to trans-femoral amputees on the basis of center of pressure.

National Magnetic Resonance Research Centre

Turkey (Remote)

Research Assistant

August-2018 to January-2019

- Investigation of tissue perfusion changes in patients undergoing carotid stenting using IVIM imaging.

National Center of Artificial Intelligence

Peshawar, Pakistan

Research Associate

January-2019 to May-2021

The sole purpose of this project is to find solutions to common healthcare problems through machine learning.

Responsibilities

- | | |
|---|---|
| <ul style="list-style-type: none"> • Conduct literature reviews • Collect and analyze Biomedical Signals & Images • Development of Biomedical Signals & Images database • Development of conventional/deep learning algorithm/models • Development of embedded system for AI Algorithm deployment • Prepare materials for submission to granting agencies and foundations | <ul style="list-style-type: none"> • Provide ready access to all experimental data for the faculty researcher and/or supervisor • Request or acquire equipment or supplies necessary for the project • Manage and respond to project related email • Mentor graduate students • Prepare progress reports for the PI and funding agency • Prepare other articles, reports, and presentations |
|---|---|

Certifications

-
- | | |
|--|--|
| <ul style="list-style-type: none"> • Deep learning Specialization • The Data Scientist's Toolbox • Data Analysis and Statistical Inference • Microsoft Certified MS Office Specialist • One day workshop on Robotics • 3rd IAPR summer school on document analysis | <ul style="list-style-type: none"> • Neural Networks and Deep Learning • Improving Deep Neural Networks: Hyperparameter tuning, Regularization and Optimization • Structuring Machine Learning Projects • Convolutional Neural Networks • Sequence Models |
|--|--|

Startup

-
- | | |
|---|-----------------------------|
| <ul style="list-style-type: none"> • Team lead at AI Thalascreeen. | <p>Sep-2019 to May 2021</p> |
|---|-----------------------------|

Technical Summary

Expreience: Vivitro mock circulatory loop, Bose biaxial testing, Confocal Imaging, Optical Coherence tomography, Narishige micropipette system, PreSens Precision Sensing system, Openfoam, Ansys Fluent, CAD/CAM, Robotics, Sensors and Instrumentation, 3D Printing, Laser engraving and cutting.

Programming languages: Matlab, Python (Tensorflow, Keras, Matplotlib, Numpy etc.), C++, Bash Scripting.